Lightweight Thermal-to-Visible Face Recognition on the Edge: Adaptation, Forgetting, and Quantization on a License-Free Benchmark

Ronit Mittal, Deepak Kumar Jain

***Abstract*—Compact face recognition (FR) networks such as EdgeFace approach state-of-the-art accuracy on visible-light benchmarks with fewer than two million parameters, yet the cameras that edge systems rely on at night or in harsh illumination are increasingly thermal, and these networks are rarely evaluated in that band. We present a systematic study of lightweight thermal-to-visible heterogeneous face recognition (HFR) under three constraints that matter on edge devices: few trainable parameters, no loss of visible-light accuracy, and 8-bit integer (INT8) deployment. To keep the study reproducible without data-access agreements, we build a subject-disjoint, five-fold protocol on the openly licensed SpeakingFaces corpus. A pretrained EdgeFace-XS model reaches 100% visible-to-visible but only 34.0% thermal-to-visible rank-1 accuracy on this protocol. We compare eight adaptation strategies over 15 runs each: full fine-tuning, three shared parameter-efficient methods (BitFit, LayerNorm tuning, and LoRA adapted to factorized layers), and four modality-specific methods, including a rank-space modulation (RSM) of the factorized layers that we introduce. Three findings emerge. First, every method that shares parameters with the visible path forgets 0.8–2.0 percentage points (pp) of LFW accuracy, whereas modality-specific LayerNorm matches the rank-1 accuracy of shared LayerNorm tuning (92.1% vs. 92.3%) with zero forgetting by construction, at a cost of 6.5 pp in true accept rate at a false accept rate of 0.1%. Second, dynamic INT8 quantization is strongly modality-asymmetric: it costs at most 0.3 pp of visible-to-visible but 2.2–13.7 pp of thermal-to-visible rank-1 accuracy, and the loss grows as fewer parameters are adapted. Third, the loss originates in the first three of the four stages, mainly the third: quantizing only the last stage, which holds 51% of the linear-layer weights, stays within 1.2 pp of FP32 for every method and is the fastest quantized configuration on an x86 CPU, where full dynamic INT8 is slower than FP32.**

***Index Terms*—Heterogeneous face recognition, thermal imaging, efficient neural networks, parameter-efficient fine-tuning, catastrophic forgetting, post-training quantization, mixed precision, edge computing.**

I. Introduction

This section motivates thermal-to-visible face recognition on edge devices, states the open questions and the novelty of this study, and lists our contributions; Sections II–VII then cover related work, preliminaries, methodology, experiments, discussion and conclusion.

R. Mittal is with the University of Petroleum and Energy Studies (UPES), Dehradun, India (e-mail: ronitmittal0@gmail.com, Ronit.28010@stu.upes.ac.in).

D. K. Jain is with the Dalian University of Technology, Dalian, China (e-mail: dkj@dlut.edu.cn).

Code, configurations and per-run results: https://github.com/INFUNDER/Face-Recognition-Research.

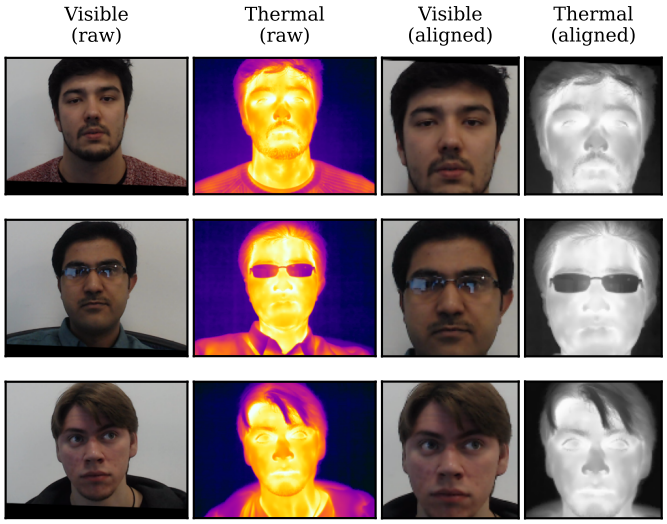


Fig. 1. Registered visible and thermal frames from SpeakingFaces [6] (CC BY 4.0) and the 112 × 112 crops produced by our alignment pipeline (Section V-A). Thermal frames are rendered in false color by the dataset and converted to gray scale for recognition. Eyeglasses are opaque at thermal wavelengths (middle row), one of the reasons thermal-to-visible matching is hard.

Face recognition (FR) has moved from the data center to the edge. Compact networks such as MobileFaceNets [1] and EdgeFace [2], the latter combining the hybrid CNN–Transformer EdgeNeXt [3] with low-rank factorized linear layers, achieve high verification accuracy with a few million parameters and a few hundred million floating-point operations, and efficient-FR challenges [4] have made such models a mature research topic. Their evaluation, however, remains almost exclusively visible-light. Many of the places where an edge FR system has to work are not well lit: building access at night, in-cabin driver monitoring and perimeter surveillance increasingly use long-wave infrared (thermal) cameras, which are insensitive to ambient illumination and harder to deceive with printed photographs [5]. Enrolment, on the other hand, almost always happens with ordinary visible-light photographs.

Matching a thermal probe against a visible gallery is a heterogeneous face recognition (HFR) problem, and a difficult one (Fig. 1). A thermal camera records emitted rather than reflected radiation, so texture, hair and eye regions look fundamentally different, and eyeglasses are opaque.

A. Open Questions for Lightweight HFR

HFR has been studied extensively for large networks [7]–[10], but three questions that decide whether it is practical on an edge device remain open for lightweight models:

1) *How should a compact model be adapted to the thermal band?* Storing a fully fine-tuned copy per modality doubles the memory footprint, and parameter-efficient fine-tuning (PEFT) methods [11]–[13] were designed for much larger networks than a 1.8 M-parameter face model.
2) *What happens to the visible-light task?* A deployed model usually has to serve visible probes as well, and adapting it to thermal data can cause catastrophic forgetting [14], [15].
3) *What does quantization do to cross-modal matching?* Edge deployment nearly always involves 8-bit integer (INT8) arithmetic [16], [17], yet quantized FR models are validated on visible-light benchmarks only.

A practical obstacle to answering these questions is data access. The standard cross-spectral face benchmarks, such as CASIA NIR-VIS 2.0 [18] and the ARL Visible-Thermal Face dataset [19], are distributed under individual license agreements, which makes it hard for many groups to reproduce or extend published results.

B. Novelty

We study all three questions jointly, on a benchmark that anyone can rebuild. The benchmark uses only SpeakingFaces [6], a corpus of synchronized and spatially registered thermal and visible video of 142 subjects released under the Creative Commons Attribution 4.0 license, which can be downloaded without a request. On it, we measure thermal accuracy, visible-light forgetting and quantized accuracy of the *same* models, which reveals interactions that separate evaluations miss: the methods that forget nothing are the ones that quantization hurts most, and the quantization loss is concentrated in one stage of the network. We also introduce a rank-space modulation (RSM) that exploits the factorized layers of EdgeFace, and show that it is most useful as a complement to normalization-based adaptation under quantization.

C. Contributions

1) **A reproducible, license-free benchmark** for lightweight thermal-to-visible FR: subject-disjoint five-fold cross-validation with fixed gallery and probe definitions, and a pipeline that downloads only the frames it needs (about 3 GB instead of the 1.2 TB release) and aligns both modalities with a single geometric transform.
2) **A controlled comparison of eight adaptation strategies** for EdgeFace over five folds and three seeds that measures thermal accuracy and visible-light forgetting jointly. It includes RSM, introduced here, and shows that normalization parameters, rather than the factorized subspaces, carry most of the modality-specific information.
3) **Evidence that INT8 quantization is modality-asymmetric for HFR**: it leaves visible matching essentially untouched but costs up to 13.7 pp of thermal rank-1 accuracy, more so for methods that adapt fewer parameters, by eroding the margin between genuine and impostor scores.
4) **A stage-wise sensitivity analysis** that traces this loss to the third of the four stages and leads to a mixed-precision configuration within 1.2 pp of FP32 for every method, together with CPU latency measurements showing that dynamic INT8 does not accelerate these low-rank models on x86 processors.

II. Related Work

This section reviews efficient face recognition, heterogeneous and thermal face recognition, parameter-efficient adaptation and forgetting, and quantization, and places our study relative to each.

A. Efficient Face Recognition

MobileFaceNets [1] adapted mobile CNN design to face embeddings, and the Lightweight Face Recognition Challenge [4] established common compute budgets for the task. EdgeFace [2] builds on EdgeNeXt [3], which combines convolutional encoders with split depth-wise transposed attention, and replaces each linear layer by a low-rank factorization (LoRaLin), reducing parameters and computation with little loss in accuracy. These models are trained with margin-based softmax losses such as ArcFace [20] on large visible-light datasets and evaluated on visible-light verification benchmarks such as LFW [21]. We ask how such a model behaves, and how it should be adapted, when the probe is thermal.

B. Heterogeneous and Thermal Face Recognition

Early thermal FR relied on hand-crafted local descriptors [5]. Deep HFR methods learn a mapping from thermal to visible features [7], learn modality-invariant representations with light-weight backbones [8], synthesize paired training data with generative models [9], or prepend a small modality-specific network to a frozen visible-light FR model, as in the Prepended Domain Transformer [10]. The last design keeps the visible path intact and is closest in spirit to the modality-specific methods studied here. Most of these works use large backbones and evaluate in floating point; edge constraints are rarely considered. Benchmarks with paired thermal and visible faces include ARL-VTF [19], the Tufts Face Database [22] and SpeakingFaces [6]; only the last is released under an open license that requires no data-access request.

C. Parameter-Efficient Adaptation and Forgetting

Adapters [11] insert small trainable modules into a frozen network, BitFit [12] tunes only bias terms, and LoRA [13] learns low-rank weight updates. Normalization parameters alone are surprisingly expressive [23], and re-estimating or separating normalization statistics per domain is an established tool for domain adaptation [24], [25]. Fine-tuning on a new

domain can degrade performance on the original one [14]; regularization-based methods such as Learning without Forgetting [15] mitigate this, whereas modality-specific parameters avoid it by construction (Proposition 1).

D. Quantization

Integer quantization [16] is the standard way to reduce the size and latency of networks on edge hardware [17], [26]–[28]. Post-training dynamic quantization stores weights in INT8 and quantizes activations on the fly, requiring no calibration data. Mixed-precision methods assign different bit widths to layers according to their sensitivity [29]. To the best of our knowledge, the interaction between quantization and cross-modal matching in lightweight HFR has not been characterized.

III. Preliminaries

This section defines the HFR task, the factorized layers and normalization of EdgeFace, and the dynamic INT8 quantization that the rest of the paper builds on.

A. Thermal-to-Visible Face Recognition

An FR network f maps an aligned face image to an embedding, and two faces are compared by the cosine similarity of their L2-normalized embeddings. In HFR, the gallery contains visible images $\boldsymbol{I}^{\text{vis}}$ and the probes are thermal images $\boldsymbol{I}^{\text{thr}}$ of the same identities. A deployed system must usually also match visible probes, so we evaluate both thermal-to-visible and visible-to-visible matching, and the modality $m \in \{\text{vis}, \text{thr}\}$ of every input is known from the camera that captured it.

B. Factorized Layers of EdgeFace

EdgeFace-XS ($\gamma = 0.6$) [2] has 1.77 M parameters, four EdgeNeXt stages, and 43 factorized linear layers

$$\boldsymbol{y} = \boldsymbol{U}\boldsymbol{V}\boldsymbol{x} + \boldsymbol{b}, \qquad \boldsymbol{U} \in \mathbb{R}^{d_o \times r},\ \boldsymbol{V} \in \mathbb{R}^{r \times d_i}, \tag{1}$$

whose rank r is a fraction γ of $\min(d_i, d_o)$. Each layer thus passes its input through an r-dimensional bottleneck $\boldsymbol{h} = \boldsymbol{V}\boldsymbol{x}$. Normalization is performed by 26 LayerNorms (21 channel-last and 5 channel-first),

$$\mathrm{LN}(\boldsymbol{x}) = \boldsymbol{\gamma} \odot \frac{\boldsymbol{x} - \mu(\boldsymbol{x})}{\sigma(\boldsymbol{x})} + \boldsymbol{\beta}, \tag{2}$$

with 5,080 affine parameters in total. Stages 1–4 hold 1.2%, 5.7%, 37.0% and 50.9% of the 1.57 M linear-layer weights, respectively, and the embedding head holds the remaining 5.2%.

C. Dynamic INT8 Quantization

We use PyTorch post-training dynamic quantization of all linear layers (fbgemm backend), the standard INT8 path for transformer-style models on CPUs. Weights are quantized per tensor and symmetrically,

$$w_q = \mathrm{clamp}\big(\lfloor w/\Delta_w \rceil, -128, 127\big), \tag{3}$$

with $\Delta_w = \max|w|/127.5$, and every input activation tensor is quantized at run time, asymmetrically, to the reduced range $[0, 127]$ that the x86 kernels use to avoid overflow,

$$x_q = \mathrm{clamp}\big(\lfloor x/\Delta_x \rceil + z, 0, 127\big), \tag{4}$$

with $\Delta_x = (x_{\max} - x_{\min})/127$, zero point $z = \lfloor -x_{\min}/\Delta_x \rceil$, and $x_{\min} \le 0 \le x_{\max}$ the tensor's extremes. Convolutions and normalization layers remain in FP32. Because Δ_x is set by the extremes of the whole tensor, a few large activations coarsen the quantization of all others.

IV. Methodology

This section describes the shared and modality-specific adaptation strategies, their properties, and the mixed-precision quantization we evaluate. All methods start from the public EdgeFace-XS checkpoint and are trained on mixed thermal and visible images with an ArcFace head that is discarded afterwards.

A. Shared Adaptation

These methods update parameters that both modalities use.

- *Full fine-tuning* trains all 1.77 M weights.
- *BitFit* [12] trains the 15,400 bias parameters.
- *LN tuning* trains the LayerNorm affine parameters $\boldsymbol{\gamma}, \boldsymbol{\beta}$ of (2) (5,080 parameters).
- *LoRA* [13] for factorized layers, defined next (20,480 parameters).

LoRA adds a trainable low-rank update $\boldsymbol{B}\boldsymbol{A}$ to a frozen weight. For a factorized layer this would add a third matrix product, so we implement it as a rank expansion:

$$\boldsymbol{y} = \begin{bmatrix}\boldsymbol{U} & \boldsymbol{B}\end{bmatrix} \begin{bmatrix}\boldsymbol{V} \\ \boldsymbol{A}\end{bmatrix} \boldsymbol{x} + \boldsymbol{b} = \boldsymbol{U}\boldsymbol{V}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}, \tag{5}$$

with $\boldsymbol{A} \in \mathbb{R}^{k \times d_i}$ randomly initialized, $\boldsymbol{B} \in \mathbb{R}^{d_o \times k}$ initialized to zero, only $\boldsymbol{A}, \boldsymbol{B}$ trained, and $k = 1$. At initialization the model is unchanged, and after training the layer is again of the form (1) with rank $r + k$, so quantized export needs no special handling.

B. Modality-Specific Adaptation

These methods keep one copy of a small set of parameters per modality $m \in \{\text{vis}, \text{thr}\}$, freeze the backbone, and fix the visible copy at its pretrained value (Fig. 2).

Modality-specific LayerNorm (msLN) follows domain-specific normalization [25]: every LayerNorm (2) keeps one pair of affine parameters per modality,

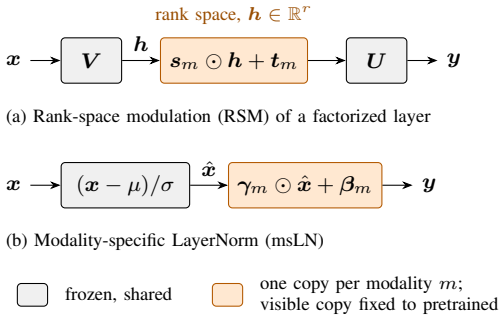
$$\mathrm{LN}_m(\boldsymbol{x}) = \boldsymbol{\gamma}_m \odot \frac{\boldsymbol{x} - \mu(\boldsymbol{x})}{\sigma(\boldsymbol{x})} + \boldsymbol{\beta}_m, \tag{6}$$

which adds 5,080 thermal parameters.

Rank-space modulation (RSM), which we introduce, exploits the bottleneck of the factorized layers (1), a natural, low-dimensional place to inject modality information:

$$\boldsymbol{y} = \boldsymbol{U}\big(\boldsymbol{s}_m \odot \boldsymbol{V}\boldsymbol{x} + \boldsymbol{t}_m\big) + \boldsymbol{b}, \tag{7}$$

with per-modality scale $\boldsymbol{s}_m \in \mathbb{R}^r$ (initialized to ones) and shift $\boldsymbol{t}_m \in \mathbb{R}^r$ (initialized to zeros). We evaluate the scale only (*RSM-s*, 2,653 thermal parameters), scale and shift (*RSM*, 5,306), and RSM combined with msLN (*RSM+msLN*, 10,386).



(a) Rank-space modulation (RSM) of a factorized layer

(b) Modality-specific LayerNorm (msLN)

Fig. 2. Modality-specific adaptation. Gray blocks are pretrained and shared; orange blocks hold one parameter vector per modality, selected by the modality of the input. Fixing the visible copy to its pretrained value makes the visible path identical to the pretrained network (Proposition 1), and both operations fold into adjacent weights (Proposition 2).

C. Properties

Two short propositions state what the modality-specific design guarantees.

Proposition 1 (Exact preservation of the visible path): If the visible copies are fixed to $\boldsymbol{s}_{\text{vis}} = \mathbf{1}$, $\boldsymbol{t}_{\text{vis}} = \mathbf{0}$, $\boldsymbol{\gamma}_{\text{vis}} = \boldsymbol{\gamma}$ and $\boldsymbol{\beta}_{\text{vis}} = \boldsymbol{\beta}$, and the backbone is frozen, then for every visible input the adapted network computes exactly the same embedding as the pretrained network. Hence every visible-light metric, including LFW accuracy, is unchanged.

Proof: With these values, (7) reduces to (1) and (6) to (2); all other operations are the frozen pretrained ones. ∎

Proposition 2 (No inference overhead): For a fixed modality m, RSM and msLN fold into the pretrained parameters: (7) equals a factorized layer (1) with $\boldsymbol{U}' = \boldsymbol{U}\,\mathrm{diag}(\boldsymbol{s}_m)$ and $\boldsymbol{b}' = \boldsymbol{U}\boldsymbol{t}_m + \boldsymbol{b}$, and (6) is a LayerNorm with parameters $(\boldsymbol{\gamma}_m, \boldsymbol{\beta}_m)$.

Proof: $\boldsymbol{U}(\boldsymbol{s}_m \odot \boldsymbol{V}\boldsymbol{x} + \boldsymbol{t}_m) + \boldsymbol{b} = \boldsymbol{U}\,\mathrm{diag}(\boldsymbol{s}_m)\boldsymbol{V}\boldsymbol{x} + (\boldsymbol{U}\boldsymbol{t}_m + \boldsymbol{b})$. The statement for msLN is immediate. ∎

Remark 1: A camera-specific deployment therefore stores one pretrained network plus 2.7–10.4 k thermal parameters, and runs either modality at exactly the cost of the original model. Shared methods offer no such guarantee: any update of a shared parameter changes the visible path as well.

D. Mixed-Precision Quantization

Besides quantizing all linear layers with (3)–(4), we quantize the linear layers of only a subset of the four stages and keep the rest, including the embedding head, in FP32. Quantizing one stage at a time measures the sensitivity of each stage, and quantizing complementary subsets gives deployable mixed-precision configurations.

V. Experiments

This section describes the benchmark, metrics and training setup, and then reports the thermal-to-visible gap, adaptation and forgetting, the effect of pose, quantization, score distributions, stage-wise sensitivity, and size and latency.

A. Dataset and Protocol

SpeakingFaces [6] contains synchronized thermal and visible video of 142 subjects reading commands, recorded in two trials per subject at nine head positions, with the visible stream registered pixel by pixel to the 464×348 thermal stream. One subject (ID 2) has no image data in the public release, leaving 141 subjects. For every subject, trial and position, we take frames 100, 300, 500 and 700 of the 900-frame clip, which yields 72 thermal–visible pairs per subject. Because the archives total about 1.2 TB, our pipeline reads only the required files from the remote archives with HTTP range requests. Table I summarizes the data, and Fig. 3 shows the nine head positions.

TABLE I
Benchmark statistics.

Subjects with image data	141
Trials × head positions × frames per subject	$2 \times 9 \times 4$
Thermal–visible pairs (aligned / total)	10,101 / 10,152
Training subjects per fold	112–113
Test subjects (gallery templates) per fold	28–29
Thermal probes per fold	1,000–1,039
Probe–template pairs per fold	≈28,000
Downloaded data	≈3 GB
License	CC BY 4.0

1) Alignment: We detect five facial landmarks with MTCNN [30] on the visible frame, estimate a similarity transform to the standard 112×112 ArcFace template, and apply the *same* transform to the registered thermal frame. Landmark detection on thermal images is unreliable, and the shared transform guarantees that both crops cover exactly the same facial region. Detection failed for 51 of the 10,152 pairs (0.5%), which are discarded. The false-color thermal rendering is converted to gray scale and replicated over three channels.

2) Five-Fold Subject-Disjoint Protocol: Subjects are assigned to five folds by subject ID (ID mod 5); for each fold, the remaining subjects form the training set. The *gallery* contains visible images of trial 1 at the frontal position, averaged into one L2-normalized template per subject. The *probes* are thermal images of trial 2 at all nine positions; trial 2 is a separate recording, so probes never share frames with the gallery. As a sanity check, the same protocol is also run with visible probes (VIS→VIS).

B. Evaluation Metrics

- **Rank-1**: closed-set identification accuracy of thermal probes against the visible templates.
- **TAR@FAR** and **EER**: verification over all probe–template pairs, reported as the true accept rate (TAR) at a false accept rate (FAR) of 10^{-3} and the equal error rate.
- **Forgetting**: LFW [21] verification accuracy of the adapted model's visible path, using the standard 10-fold protocol (pretrained: 99.73%).

Every method is trained on each of the five training sets with three seeds, and we report the mean ± standard deviation over the 15 runs. Because the folds differ in difficulty, the standard deviation mostly reflects fold-to-fold variation; when two methods are compared, we also report paired differences over the same fold and seed.

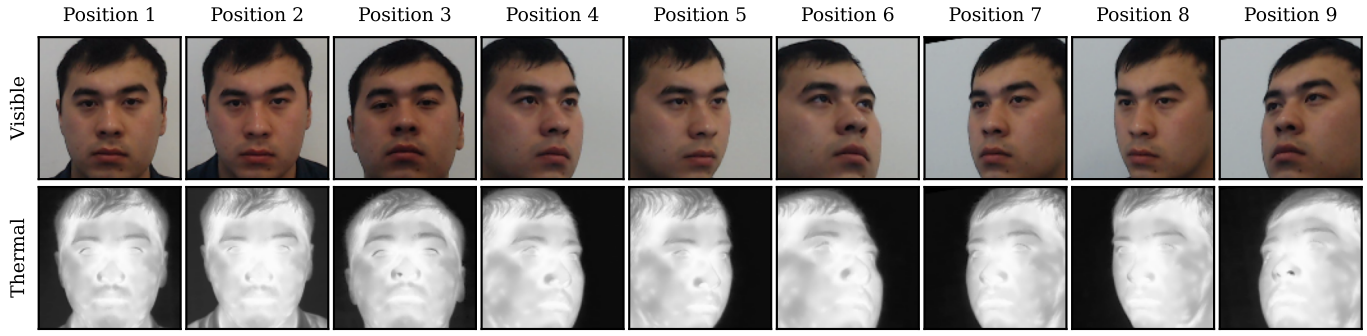


Fig. 3. Aligned visible (top) and thermal (bottom) crops of one SpeakingFaces subject at the nine head positions (trial 2). Position 1 is frontal; the gallery uses frontal visible images of trial 1, and thermal probes are taken from all positions of trial 2.

C. Implementation Details

All models are trained for 20 epochs with batch size 128, AdamW, a one-cycle learning-rate schedule with 10% warm-up, and the ArcFace loss [20] (scale 64, margin 0.5). The ArcFace head uses a learning rate of 10^{-2}; full fine-tuning uses 10^{-4} with weight decay 0.05; BitFit, LN tuning, LoRA and msLN use 10^{-3}; RSM vectors use 10^{-2}; the adapted parameters of the parameter-efficient methods have no weight decay. Horizontal flipping is the only augmentation. For each method, the learning rate was chosen from two candidates on a development split (training subjects 1–80, validation subjects 81–100) with the same budget for every method. Training one model takes 3–5 minutes on one NVIDIA H100 GPU. INT8 models are evaluated on CPU, since the dynamic INT8 kernels are CPU-only. Latency is measured with PyTorch 2.4 on an Intel Xeon Platinum 8470 CPU for single 112×112 images, as the median of 200 runs after 20 warm-up runs, with one and four threads.

D. The Thermal-to-Visible Gap

Without adaptation, EdgeFace-XS reaches 100% VIS→VIS rank-1 accuracy on the benchmark but only $34.0 \pm 6.7\%$ thermal-to-visible rank-1 accuracy, with a TAR@FAR = 0.1% of 6.3% and an EER of 24.7% (Table II). The network has excellent visible-light features but transfers almost nothing to the thermal band, which motivates adaptation.

E. Adaptation and Forgetting

All methods close most of the gap (Table II). Full fine-tuning is the most accurate, with 95.6% rank-1 accuracy and 83.8% TAR@0.1%, and it outperforms LoRA in all 15 paired runs (+2.0 pp rank-1, +6.0 pp TAR). It also forgets: LFW accuracy falls from 99.73% to 98.52%. Every shared method forgets, between 0.77 pp (LoRA) and 1.95 pp (BitFit). Because LFW is close to saturation, these drops correspond to a 3.9- to 8.2-fold increase in LFW verification errors (from 0.27% to 1.04–2.22%).

The modality-specific methods leave LFW at exactly 99.73%, as Proposition 1 guarantees. Among them, msLN is the strongest. It matches shared LN tuning, which trains the same number of parameters, in rank-1 accuracy (paired difference -0.26 ± 2.58 pp), but is 6.5 pp lower in TAR@0.1%

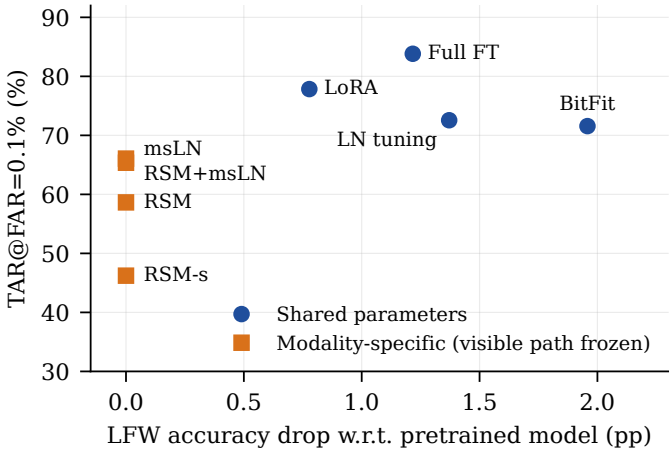


Fig. 4. Thermal-to-visible verification accuracy versus visible-light forgetting (FP32, mean of 15 runs). Modality-specific methods have zero forgetting by construction; shared methods trade LFW accuracy for thermal accuracy.

(lower in 14 of 15 runs): being able to also adapt the normalization of the visible path helps verification at strict operating points, at the price of forgetting. RSM is less effective than msLN on its own. The scale-only variant reaches 84.1% rank-1 accuracy; adding the shift improves every run (+4.3 pp rank-1, +12.4 pp TAR, 15/15 runs), but RSM still trails msLN by 3.7 pp rank-1 accuracy in all 15 runs at the same parameter budget. Combining RSM with msLN does not improve on msLN in FP32 (-0.8 pp rank-1, -0.7 pp TAR) but makes it more robust to quantization (Section V-G). Together, these results indicate that, for a compact backbone, the normalization statistics rather than the subspaces of the factorized layers carry most of the modality-specific information.

Fig. 4 summarizes the trade-off. Shared methods form an accuracy frontier that can only be reached by giving up visible accuracy, and the best zero-forgetting method lies 5.5–6.5 pp below the nearest shared methods in TAR@0.1%.

F. Effect of Head Pose

The gallery contains frontal visible images only, while thermal probes span nine head positions (Fig. 3). Fig. 5 breaks rank-1 accuracy down by probe position (seed 0, five folds). Without adaptation, accuracy varies between 27.7% (position 9) and 40.5% (position 2), but it is low everywhere, so the

TABLE II
THERMAL-TO-VISIBLE HFR ON SPEAKINGFACES (5 FOLDS × 3 SEEDS, MEAN±STD). PARAMS: TRAINABLE PARAMETERS. LFW: VERIFICATION ACCURACY OF THE VISIBLE PATH (PRETRAINED: 99.73%). MODALITY-SPECIFIC METHODS KEEP THE VISIBLE PATH IDENTICAL TO THE PRETRAINED NETWORK. THE BEST RESULTS ARE **BOLDED**.

			FP32			INT8 (all linear layers)		
Type	Method	Params	Rank-1 (%)	TAR@0.1%	EER (%)	Rank-1 (%)	TAR@0.1%	LFW (%)
–	Zero-shot	0	33.96 ±6.73	6.32 ±4.56	24.72	28.83 ±6.19	5.19 ±4.01	99.73
Shared	Full fine-tuning	1.77 M	**95.58** ±4.29	**83.83** ±5.68	**2.70**	**93.38** ±4.76	**76.70** ±8.04	98.52 ±0.18
	LoRA ($k = 1$)	20.5 k	93.58 ±3.87	77.84 ±7.38	3.64	89.10 ±5.22	64.20 ±7.17	98.96 ±0.11
	BitFit	15.4 k	92.85 ±4.86	71.57 ±9.80	3.79	89.90 ±5.24	64.35 ±9.36	97.78 ±0.16
	LN tuning	5.1 k	92.31 ±4.78	72.56 ±7.35	4.28	86.96 ±5.34	58.54 ±7.06	98.36 ±0.18
Modality-specific	msLN	5.1 k	92.05 ±2.90	66.06 ±4.30	5.15	84.66 ±4.15	50.81 ±7.83	**99.73**
	RSM+msLN (ours)	10.4 k	91.25 ±3.50	65.36 ±6.76	5.77	86.22 ±3.89	54.43 ±6.57	**99.73**
	RSM (ours)	5.3 k	88.37 ±3.84	58.64 ±9.37	6.17	82.45 ±4.94	44.51 ±9.38	**99.73**
	RSM-s (ours)	2.7 k	84.10 ±4.52	46.22 ±10.79	6.95	70.43 ±5.96	28.66 ±5.65	**99.73**

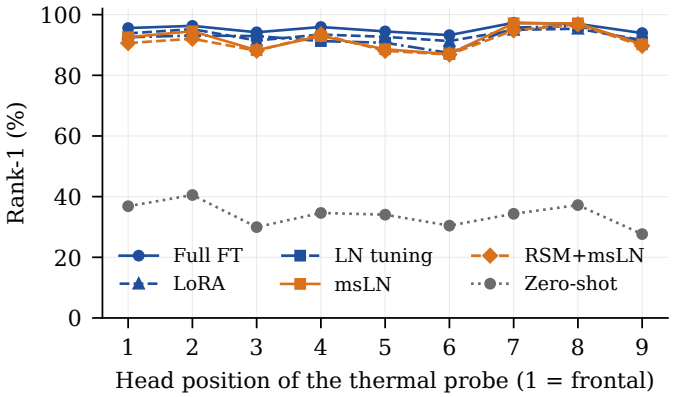


Fig. 5. Thermal-to-visible rank-1 accuracy per head position of the probe (FP32, seed 0, mean over five folds). The gallery contains frontal visible images only.

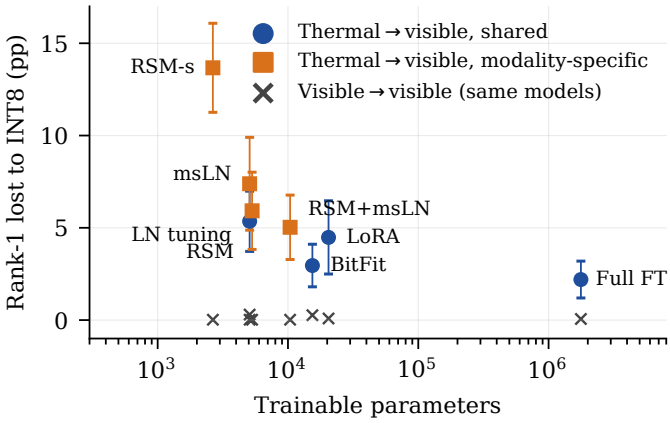


Fig. 6. Rank-1 accuracy lost to full dynamic INT8 quantization (FP32 minus INT8, mean±std of 15 paired runs) versus the number of trainable parameters. Crosses show the loss for visible probes of the same models, which never exceeds 0.3 pp.

modality gap rather than pose is the dominant difficulty. After adaptation, accuracy is nearly uniform across positions: 93.2–97.3% for full fine-tuning and 87.0–97.2% for msLN. Position 6 is the hardest for every adapted model, and positions 7 and 8 are among the easiest.

G. Quantization Is Modality-Asymmetric

Fig. 6 compares, for the same models, the rank-1 accuracy lost to INT8 quantization for thermal and for visible probes. For visible probes the loss is negligible (at most 0.3 pp). For thermal probes it ranges from 2.2 pp (full fine-tuning) to 13.7 pp (RSM-s), and even the unadapted model loses 5.1 pp. Quantization noise that is harmless for well-separated visible embeddings is thus significant for thermal embeddings, which lie much closer to the decision boundary (Section V-H).

The loss also grows as fewer parameters are adapted: 2.2 pp for full fine-tuning, 3.0–7.4 pp for the methods with 5–20 k parameters, and 13.7 pp for the 2.7 k-parameter RSM-s. A plausible explanation is that full fine-tuning can reshape the whole network so that thermal embeddings become well separated, whereas methods that modulate a frozen network rely on small, precise feature shifts that quantization noise can overwhelm. Among the modality-specific methods, adding RSM to msLN reduces the INT8 loss from 7.4 ±2.5 to 5.0 ±1.7 pp and gives the best quantized accuracy without forgetting (86.2% rank-1, 54.4% TAR@0.1%).

H. Score Distributions

Fig. 7 shows the distributions of cosine similarity between thermal probes and visible templates (seed 0, all folds pooled). Without adaptation, genuine pairs score only 0.145 on average against 0.012 for impostor pairs (decidability $d' = 1.39$). Adaptation mainly raises the genuine scores, to 0.486 for full fine-tuning ($d' = 4.11$) and 0.355 for msLN ($d' = 3.33$), while the impostor distribution stays centered at zero. The modality-specific methods therefore separate the two classes with a smaller margin, which is consistent with their lower TAR at strict FAR. INT8 quantization leaves the impostor distribution essentially unchanged (mean shift ≤ 0.015) but moves the genuine distribution toward it, by 0.030 for full fine-tuning, 0.044 for msLN and 0.049 for LoRA, reducing d' to 3.80 and 2.92 for full fine-tuning and msLN, respectively. Quantization thus erodes exactly the margin that adaptation has built, and the smaller that margin, the more identification errors it causes.

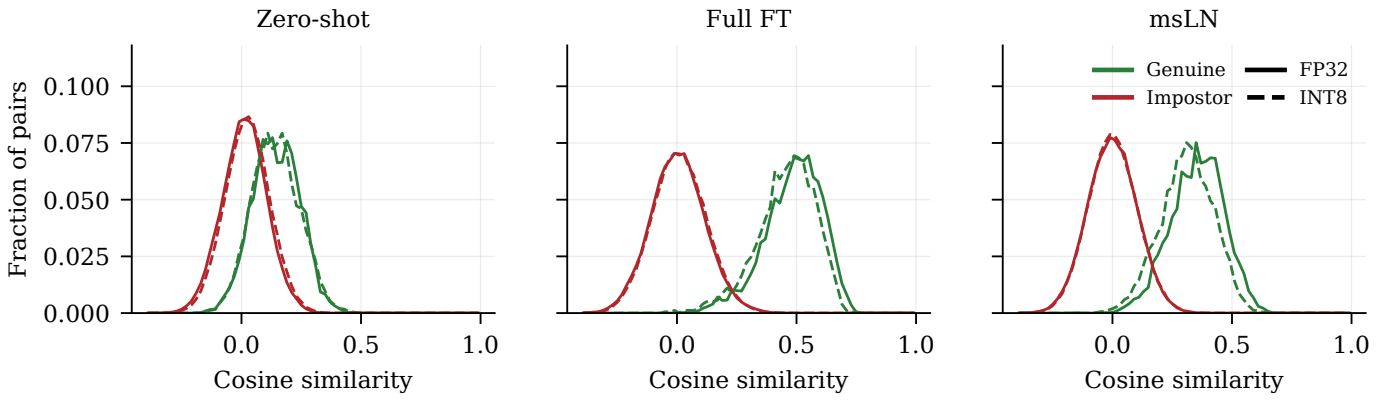


Fig. 7. Distributions of thermal-probe–visible-template cosine similarity for genuine and impostor pairs (seed 0, five folds pooled), in FP32 (solid) and with all linear layers in INT8 (dashed). Quantization shifts genuine scores toward the impostor distribution, which itself is unaffected.

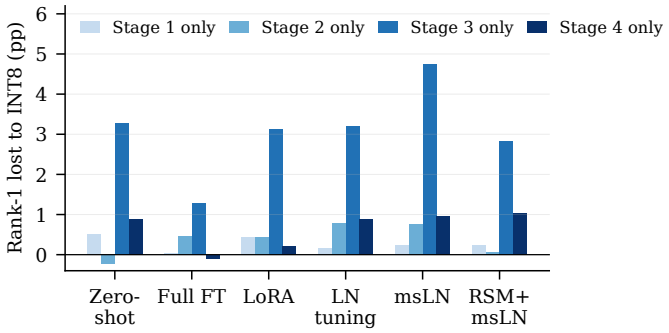


Fig. 8. Thermal-to-visible rank-1 accuracy lost when the linear layers of a single stage are quantized to INT8 and all other layers stay in FP32 (seed 0, mean over five folds). Stage 3 accounts for most of the loss of every method.

I. Where the Quantization Loss Comes From

To locate the loss, we quantize the linear layers of only some stages (Table III and Fig. 8). Quantizing stages 1–3, which hold 44% of the linear-layer weights, reproduces almost the whole INT8 loss, whereas quantizing only stage 4, which holds 51%, costs between -0.01 and 1.2 pp of rank-1 accuracy relative to FP32 for every method. Sensitivity is therefore far from proportional to parameter count: the stage with the most weights is the least sensitive.

Quantizing one stage at a time (Fig. 8; seed 0, five folds) identifies stage 3 as the main source of the loss: on its own it costs 1.3 pp of rank-1 accuracy for full fine-tuning and 2.8–4.7 pp for the parameter-efficient methods, whereas stages 1, 2 and 4 each cost about 1 pp or less. Stage 3 contains 20 of the 43 factorized layers and operates at 7×7 resolution. In five of the six models, the single-stage losses add up to less than the loss of full INT8, indicating that errors introduced in different stages compound as they propagate through the network.

J. Size and Latency

Table IV reports CPU latency and model size. On this x86 CPU, dynamic INT8 is *slower* than FP32 for both EdgeFace-XS (7.96 vs. 6.14 ms with one thread) and EdgeFace-S (9.29 vs. 7.96 ms). The factorized layers have small inner dimensions, so the cost of quantizing activations at run time and the extra kernel calls outweigh the savings in arithmetic. Quantizing only stage 4 is the fastest quantized configuration (6.53 ms, 6% slower than FP32) and reduces the model size by 33%, while keeping accuracy within 1.2 pp of FP32 (Table III). Full INT8 gives the largest size reduction ($2.8\times$), which matters when storage or memory bandwidth rather than computation is the bottleneck.

VI. Discussion

This section turns the results into deployment recommendations, reports negative results, and discusses limitations and ethical considerations.

A. Recommendations for Deployment

1) *Separate thermal model allowed:* if a second model can be stored and a small loss of visible accuracy is acceptable, full fine-tuning is the most accurate choice and the most robust to quantization.
2) *One model for both modalities:* if visible-light accuracy must not regress, modality-specific LayerNorm is a strong and simple default with 5 k extra parameters and no inference overhead (Proposition 2); adding RSM improves its quantized accuracy.
3) *Quantized deployment:* keep stages 1–3, stage 3 above all, in higher precision; quantizing only the last stage recovers almost all of the INT8 loss at a modest cost in size and is the fastest quantized configuration on x86 CPUs.
4) *Validation:* evaluate quantized models on the target modality; visible-light benchmarks hide quantization losses of up to 13.7 pp in thermal rank-1 accuracy.

B. Negative Results

In preliminary experiments on the development split (single seed), we also trained the modality-specific methods with quantization-aware training, using a fake-quantization module that reproduces PyTorch's dynamic INT8 to within 10^{-4} mean relative error per layer, and with a distillation loss that pulls each thermal embedding toward the pretrained embedding of its registered visible frame. Neither gave consistent gains beyond seed-to-seed variation, so we did not pursue them in the full protocol. We report them so that others need not repeat them, with the caveat that a single development run cannot rule out benefits.

TABLE III
MIXED-PRECISION INT8: THERMAL-TO-VISIBLE RANK-1 / TAR@0.1% (%) WHEN ONLY SOME STAGES' LINEAR LAYERS ARE QUANTIZED (5 FOLDS × 3 SEEDS). SIZES ARE FOR THE SERIALIZED EDGEFACE-XS MODEL. THE BEST QUANTIZED RESULTS ARE **BOLDED**.

	FP32 (6.85 MB)	INT8 stage 4 only (4.59 MB)	INT8 stages 1–3 only (4.95 MB)	INT8 all (2.45 MB)
Zero-shot	33.96 / 6.32	33.08 / 6.06	30.15 / 5.61	28.83 / 5.19
Full fine-tuning	95.58 / 83.83	**95.58 / 82.82**	93.53 / 77.10	93.38 / 76.70
LoRA	93.58 / 77.84	93.07 / 76.28	89.86 / 66.78	89.10 / 64.20
BitFit	92.85 / 71.57	92.55 / 70.83	90.39 / 66.22	89.90 / 64.35
LN tuning	92.31 / 72.56	91.48 / 71.02	87.74 / 60.50	86.96 / 58.54
msLN	92.05 / 66.06	90.88 / 63.98	85.85 / 52.87	84.66 / 50.81
RSM+msLN	91.25 / 65.36	90.27 / 63.23	86.89 / 57.25	86.22 / 54.43
RSM	88.37 / 58.64	87.35 / 56.21	83.37 / 47.94	82.45 / 44.51
RSM-s	84.10 / 46.22	83.05 / 44.06	71.81 / 31.27	70.43 / 28.66

TABLE IV
SERIALIZED SIZE AND MEDIAN CPU LATENCY PER 112 × 112 FACE (INTEL XEON PLATINUM 8470, PYTORCH 2.4, DYNAMIC INT8 WITH FBGEMM).

		Size	Latency (ms)	
Model	Precision	(MB)	1 thread	4 threads
XS	FP32	6.85	**6.14**	**5.23**
	INT8 stage 4	4.59	6.53	5.70
	INT8 stages 1–3	4.95	7.92	7.04
	INT8 all	**2.45**	7.96	7.24
S	FP32	14.03	**7.96**	**6.24**
	INT8 stage 4	9.28	8.10	6.59
	INT8 stages 1–3	9.99	9.66	7.90
	INT8 all	**4.89**	9.29	8.11

C. Limitations

1) *Single dataset:* 141 subjects recorded in one laboratory with one thermal camera; absolute numbers will not transfer directly to other sensors or uncontrolled conditions.
2) *Gallery size:* each test fold contains 28–29 identities, so rank-1 accuracy is optimistic compared with large-gallery deployments, which is why we also report TAR at a strict FAR.
3) *Hyper-parameter selection:* learning rates were selected on subjects 81–100, which also appear in test folds of the cross-validation; the selection grid was very small (two values per method) and identical across methods, but this is a mild form of leakage.
4) *Quantization scheme:* only post-training dynamic INT8 with per-tensor scales was studied; static, per-channel or lower-bit quantization may behave differently.
5) *Hardware:* latency was measured on a server-class x86 CPU; ARM processors and neural accelerators use different INT8 kernels and may show different speed-ups.

D. Ethical Considerations

Face recognition, and thermal surveillance in particular, raises privacy concerns. This work uses only a public dataset whose participants consented to the release of their recordings under an open license, and the sample images shown are taken from that release. The adaptation methods studied here do not increase the capabilities of FR systems beyond matching across spectra, and the benchmark is intended to support the transparent evaluation of such systems.

VII. CONCLUSION

We studied how a face recognition model with fewer than two million parameters can be adapted to thermal-to-visible matching under edge constraints, using a reproducible benchmark built only on openly licensed data. Adaptation closes most of the 66 pp rank-1 gap, but every method that shares parameters with the visible path forgets visible-light accuracy, while modality-specific methods that keep the visible path intact trail by about 6 pp in TAR at FAR = 0.1%. Among the latter, normalization-based adaptation is the most effective, and the rank-space modulation we introduce mainly helps under quantization. INT8 quantization hurts thermal matching much more than visible matching, increasingly so for methods that adapt fewer parameters, by eroding the margin between genuine and impostor scores; this loss originates mainly in the third stage, so a simple mixed-precision configuration recovers it. Future work will extend the benchmark to other openly licensed thermal datasets and measure latency on embedded ARM hardware.

DATA AVAILABILITY

SpeakingFaces is available under the CC BY 4.0 license at https://huggingface.co/datasets/issai/Speaking_Faces. The download and alignment pipeline, training and evaluation code, experiment configurations and per-run results are available at https://github.com/INFUNDER/Face-Recognition-Research.

ACKNOWLEDGMENT

The authors thank the creators of SpeakingFaces for releasing the dataset under an open license, and acknowledge the high-performance computing facility used for the experiments.

REFERENCES

[1] S. Chen, Y. Liu, X. Gao, and Z. Han, "MobileFaceNets: Efficient CNNs for accurate real-time face verification on mobile devices," in *Proc. Chinese Conference on Biometric Recognition (CCBR), Lecture Notes in Computer Science*, 2018, pp. 428–438.
[2] A. George, C. Ecabert, H. Otroshi Shahreza, K. Kotwal, and S. Marcel, "EdgeFace: Efficient face recognition model for edge devices," *IEEE Transactions on Biometrics, Behavior, and Identity Science*, vol. 6, no. 2, pp. 158–168, 2024.

[3] M. Maaz, A. Shaker, H. Cholakkal, S. Khan, S. W. Zamir, R. M. Anwer, and F. S. Khan, "EdgeNeXt: Efficiently amalgamated CNN-transformer architecture for mobile vision applications," in *Computer Vision – ECCV 2022 Workshops, Lecture Notes in Computer Science*. Springer, 2023, pp. 3–20.

[4] J. Deng, J. Guo, D. Zhang, Y. Deng, X. Lu, and S. Shi, "Lightweight face recognition challenge," in *Proc. IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, 2019, pp. 2638–2646.

[5] G. Hermosilla, J. Ruiz-del Solar, R. Verschae, and M. Correa, "A comparative study of thermal face recognition methods in unconstrained environments," *Pattern Recognition*, vol. 45, no. 7, pp. 2445–2459, 2012.

[6] M. Abdrakhmanova, A. Kuzdeuov, S. Jarju, Y. Khassanov, M. Lewis, and H. A. Varol, "SpeakingFaces: A large-scale multimodal dataset of voice commands with visual and thermal video streams," *Sensors*, vol. 21, no. 10, p. 3465, 2021.

[7] M. S. Sarfraz and R. Stiefelhagen, "Deep perceptual mapping for cross-modal face recognition," *International Journal of Computer Vision*, vol. 122, no. 3, pp. 426–438, 2017.

[8] X. Wu, R. He, Z. Sun, and T. Tan, "A light CNN for deep face representation with noisy labels," *IEEE Transactions on Information Forensics and Security*, vol. 13, no. 11, pp. 2884–2896, 2018.

[9] C. Fu, X. Wu, Y. Hu, H. Huang, and R. He, "Dual variational generation for low shot heterogeneous face recognition," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[10] A. George, A. Mohammadi, and S. Marcel, "Prepended domain transformer: Heterogeneous face recognition without bells and whistles," *IEEE Transactions on Information Forensics and Security*, vol. 18, pp. 133–146, 2023.

[11] N. Houlsby, A. Giurgiu, S. Jastrzebski, B. Morrone, Q. de Laroussilhe, A. Gesmundo, M. Attariyan, and S. Gelly, "Parameter-efficient transfer learning for NLP," in *Proc. International Conference on Machine Learning (ICML), PMLR vol. 97*, 2019, pp. 2790–2799.

[12] E. Ben Zaken, Y. Goldberg, and S. Ravfogel, "BitFit: Simple parameter-efficient fine-tuning for transformer-based masked language-models," in *Proc. Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022, pp. 1–9.

[13] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in *Proc. International Conference on Learning Representations (ICLR)*, 2022.

[14] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska *et al.*, "Overcoming catastrophic forgetting in neural networks," *Proceedings of the National Academy of Sciences*, vol. 114, no. 13, pp. 3521–3526, 2017.

[15] Z. Li and D. Hoiem, "Learning without forgetting," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 40, no. 12, pp. 2935–2947, 2018.

[16] B. Jacob, S. Kligys, B. Chen, M. Zhu, M. Tang, A. Howard, H. Adam, and D. Kalenichenko, "Quantization and training of neural networks for efficient integer-arithmetic-only inference," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 2704–2713.

[17] R. Krishnamoorthi, "Quantizing deep convolutional networks for efficient inference: A whitepaper," arXiv:1806.08342, 2018.

[18] S. Z. Li, D. Yi, Z. Lei, and S. Liao, "The CASIA NIR-VIS 2.0 face database," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2013, pp. 348–353.

[19] D. Poster, M. Thielke, R. Nguyen, S. Rajaraman, X. Di, C. N. Fondje, V. M. Patel, N. J. Short, B. S. Riggan, N. M. Nasrabadi, and S. Hu, "A large-scale, time-synchronized visible and thermal face dataset," in *Proc. IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2021, pp. 1558–1567.

[20] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive angular margin loss for deep face recognition," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 4685–4694.

[21] G. B. Huang, M. Ramesh, T. Berg, and E. Learned-Miller, "Labeled faces in the wild: A database for studying face recognition in unconstrained environments," University of Massachusetts, Amherst, Tech. Rep. 07-49, Oct. 2007.

[22] K. Panetta, Q. Wan, S. Agaian, S. Rajeev, S. Kamath, R. Rajendran, S. P. Rao, A. Kaszowska, H. A. Taylor, A. Samani, and X. Yuan, "A comprehensive database for benchmarking imaging systems," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 42, no. 3, pp. 509–520, 2020.

[23] J. Frankle, D. J. Schwab, and A. S. Morcos, "Training BatchNorm and only BatchNorm: On the expressive power of random features in CNNs," in *Proc. International Conference on Learning Representations (ICLR)*, 2021.

[24] Y. Li, N. Wang, J. Shi, X. Hou, and J. Liu, "Adaptive batch normalization for practical domain adaptation," *Pattern Recognition*, vol. 80, pp. 109–117, 2018.

[25] W.-G. Chang, T. You, S. Seo, S. Kwak, and B. Han, "Domain-specific batch normalization for unsupervised domain adaptation," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 7346–7354.

[26] M. Nagel, M. Fournarakis, R. A. Amjad, Y. Bondarenko, M. van Baalen, and T. Blankevoort, "A white paper on neural network quantization," arXiv:2106.08295, 2021.

[27] A. Gholami, S. Kim, Z. Dong, Z. Yao, M. W. Mahoney, and K. Keutzer, "A survey of quantization methods for efficient neural network inference," arXiv:2103.13630, 2021.

[28] H. Wu, P. Judd, X. Zhang, M. Isaev, and P. Micikevicius, "Integer quantization for deep learning inference: Principles and empirical evaluation," arXiv:2004.09602, 2020.

[29] Z. Dong, Z. Yao, A. Gholami, M. W. Mahoney, and K. Keutzer, "HAWQ: Hessian AWare quantization of neural networks with mixed-precision," in *Proc. IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 293–302.

[30] K. Zhang, Z. Zhang, Z. Li, and Y. Qiao, "Joint face detection and alignment using multitask cascaded convolutional networks," *IEEE Signal Processing Letters*, vol. 23, no. 10, pp. 1499–1503, 2016.